

Exact Joint ANF Leap for Raw Finite-Field Inversion

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China
xiaojianzeng@foxmail.com

Abstract

For every $n \geq 2$ and every monic irreducible polynomial of degree n over $\mathbb{F}_2$, consider inversion in the polynomial basis and the full Boolean map

$$(P, u) \longmapsto \text{adj}(P)J(Pu),$$

where P ranges over all binary matrices, including singular matrices. We prove that the ordering leap of the joint algebraic-normal-form support is exactly n . The lower bound follows from vanishing on every input of Hamming weight below n . For the matching upper bound, we exhibit n supported degree- n monomials arising from cyclic rank- $(n-1)$ partial permutation matrices. They are pairwise disjoint and together cover every operand variable and every matrix variable outside one row, leaving at most n new variables for the rest of the support.

1 Context and statement

The joint ANF leap records how many previously unseen variables a monomial must introduce in the best ordering of the support. It is an ANF analogue of support-ordering notions associated with leap complexity [1]. For raw finite-field inversion, Zhang and Zhang [2] proved the universal lower bound n and reported equality in exhaustive computations for $n = 3, 4$. The purpose of this note is to give the matching upper bound uniformly in both the dimension and the irreducible modulus. No invertibility restriction is imposed on the matrix input; indeed, the upper-bound witnesses below are singular.

Fix an integer $n \geq 2$ and a monic irreducible polynomial

$$f(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_1t + 1 \in \mathbb{F}_2[t].$$

The constant coefficient is one because f is irreducible and $n \geq 2$. Let $K = \mathbb{F}_2[t]/(f)$, put $\alpha = t \bmod f$, and identify $\mathbb{F}_2^n$ with K through the polynomial-basis isomorphism

$$\phi(x_0, \dots, x_{n-1}) = \sum_{j=0}^{n-1} x_j \alpha^j.$$

Let

$$J_{n,f}(0) = 0, \quad J_{n,f}(x) = \phi^{-1}(\phi(x)^{-1}) \quad (x \neq 0)$$

denote inversion in these coordinates.

Let $P = (p_{ab})_{0 \leq a, b < n}$ be an arbitrary matrix over $\mathbb{F}_2$ and let $u = (u_0, \dots, u_{n-1})^\top \in \mathbb{F}_2^n$. We use the adjugate orientation

$$\text{adj}(P)_{ab} = \det(P \text{ with row } b \text{ and column } a \text{ deleted}), \quad (1)$$

where signs disappear over $\mathbb{F}_2$. Define on the entire Boolean space

$$G_{n,f}(P, u) = \text{adj}(P) J_{n,f}(Pu). \quad (2)$$

Thus (2) is defined for singular as well as invertible P .

Write

$$\mathcal{V} = \{p_{ab} : 0 \leq a, b < n\} \cup \{u_b : 0 \leq b < n\}.$$

The unique multilinear vector ANF of $G = G_{n,f}$ is

$$G = \bigoplus_{A \subseteq \mathcal{V}} c_A \prod_{z \in A} z, \quad c_A = \bigoplus_{C \subseteq A} G(1_C), \quad (3)$$

where 1_C is the indicator assignment of C and $c_A \in \mathbb{F}_2^n$. Set $\mathcal{S} = \{A : c_A \neq 0\}$. For an ordering $(A_1, \dots, A_s)$ of the finite set $\mathcal{S}$, its largest leap is

$$\max_{1 \leq j \leq s} \left| A_j \setminus \bigcup_{\ell < j} A_\ell \right|,$$

and $L(G)$ is the minimum of this quantity over all orderings of $\mathcal{S}$.

Theorem 1. *For every integer $n \geq 2$ and every monic irreducible degree- n polynomial $f \in \mathbb{F}_2[t]$, the map in (2), on the full Boolean matrix domain and in the polynomial coordinate basis, satisfies*

$$L(G_{n,f}) = n.$$

2 Proof by a cyclic support prefix

We first prove the lower bound directly. The value of G is zero at every input assignment of Hamming weight less than n . If $u = 0$, this follows from $J_{n,f}(0) = 0$. If $u \neq 0$, at least one input one occurs in u , so P has at most $n - 2$ nonzero entries. Consequently $\text{rank}(P) \leq n - 2$, all of its $(n - 1)$ -minors vanish, and $\text{adj}(P) = 0$.

If $|A| < n$, every term $G(1_C)$ in (3) therefore vanishes. Hence

$$c_A = 0 \quad (|A| < n). \quad (4)$$

The map G is not zero, since $G(I, u) = J_{n,f}(u)$, so its support is nonempty. The first member of every support ordering has size at least n by (4), proving

$$L(G) \geq n. \quad (5)$$

It remains to construct a suitable supported prefix. Index all coordinates by $0, \dots, n - 1$ and fix $i = 1$. Since $f(\alpha) = 0$, division by the nonzero element α gives

$$\alpha^{-1} = \alpha^{n-1} + a_{n-1}\alpha^{n-2} + \dots + a_2\alpha + a_1. \quad (6)$$

Let $v = J_{n,f}(e_i)$, the coordinate vector of α^{-1} . If $n \geq 3$, choose $r = n - 1$; then $v_r = 1$ by (6), and $r \neq i$. If $n = 2$, irreducibility forces $f(t) = t^2 + t + 1$. Choose $r = 0$; again $v_r = 1$ and $r \neq i$. Thus in every case we have selected indices satisfying

$$r \neq i, \quad v_r = 1. \quad (7)$$

For $k = 0, \dots, n - 1$, let $\sigma_k(h) = h + k \pmod{n}$, put $d_k = \sigma_k(r)$, and define the n -element variable set

$$A_k = \{p_{h,\sigma_k(h)} : h \neq r\} \cup \{u_{\sigma_k(i)}\}. \quad (8)$$

At the assignment 1_{A_k} , the matrix $P^{(k)}$ has one nonzero entry in each row other than r , its row r is zero, and its missing column is d_k . Since $r \neq i$, multiplication by $u = e_{\sigma_k(i)}$ yields

$$P^{(k)}u = e_i. \quad (9)$$

Deleting row r and column d_k from $P^{(k)}$ leaves a permutation matrix. Every other $(n - 1)$ -minor either retains the zero row or removes the unique one from a retained row. With the orientation in (1), this gives

$$\text{adj}(P^{(k)}) = e_{d_k} e_r^\top. \quad (10)$$

Every proper subset of A_k has cardinality below n , so (4) makes all of its evaluations in (3) zero. Equations (7)–(10) then show that the exact ANF coefficient is

$$\begin{aligned} c_{A_k} &= G(1_{A_k}) = e_{d_k} e_r^\top J_{n,f}(e_i) \\ &= e_{d_k} v_r = e_{d_k} \neq 0. \end{aligned} \quad (11)$$

Thus every A_k lies in the joint support.

The sets $A_0, \dots, A_{n-1}$ are pairwise disjoint. For each fixed $h \neq r$, the entries $p_{h,\sigma_k(h)}$ exhaust row h as k varies, and the entries $u_{\sigma_k(i)}$ exhaust the operand variables. Hence

$$\bigcup_{k=0}^{n-1} A_k = \{p_{hj} : h \neq r, 0 \leq j < n\} \cup \{u_j : 0 \leq j < n\}. \quad (12)$$

Order these n supported sets first. Each introduces exactly n variables. After this prefix, (12) says that the only unseen variables are the n entries in row r of P . Append every other member of $\mathcal{S}$ in any order; none can introduce more than those n remaining variables. This ordering has largest leap at most n , so $L(G) \leq n$. Together with (5), this proves Theorem 1. $\square$

3 Comparison, checks, and limitations

The closest inspected source [2] defines the same full-space raw map and joint support invariant. It establishes $L(G) \geq n$ and gives finite certificates for $n = 3, 4$. The additional point here is the uniform cyclic family (8): its coefficients are nonzero for every allowed modulus, and its union leaves only one matrix row unseen. This supplies the general upper bound without enumerating the rest of the ANF support. The theorem is restricted to the polynomial coordinate basis in its statement; no claim is made here for arbitrary ordered bases. The literature comparison is bounded to the sources described above and is not a claim of priority.

Exact binary calculations were also used during proof development. Complete ANFs were checked for $n = 2, 3$, and the coefficients in (11) were checked for every irreducible modulus of degrees 2 through 6. These finite checks are corroborative only and are not used in the proof.

Computational and AI assistance. Exact-arithmetic programs were used for the finite checks just described and software tools were used for typesetting and build verification. An AI system assisted with proof exploration, adversarial checking, and manuscript preparation. The mathematical conclusion rests on the self-contained argument above rather than on numerical output or AI assessment; the author is responsible for the final text.

References

- [1] E. Abbe, E. Boix-Adserà, and T. Misiakiewicz. SGD learning on neural networks: Leap complexity and saddle-to-saddle dynamics. In *Proceedings of the 36th Conference on Learning Theory*, volume 195 of *Proceedings of Machine Learning Research*, pages 2552–2623. PMLR, 2023. <https://proceedings.mlr.press/v195/abbe23a.html>.
- [2] Z. Zhang and N. Zhang. Representation redundancy and structural complexity in finite-field inversion, 2026. arXiv:2609.04583v1. <https://arxiv.org/abs/2609.04583>.